DP Biology – B3.2 (HL, Plant Systems)

Root pressure and phloem

- Which adaptations help animals and plants move fluids around?
- How does transport in animals compare with transport in plants?

HL statements B3.2.17–18. Textbook pp. 321–325.



Image: part of a sunflower stem, stained section. Jon Houseman, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.2.17–18	Explain how root pressure is generated, and how phloem is adapted for translocation

Root pressure

B3.2.17



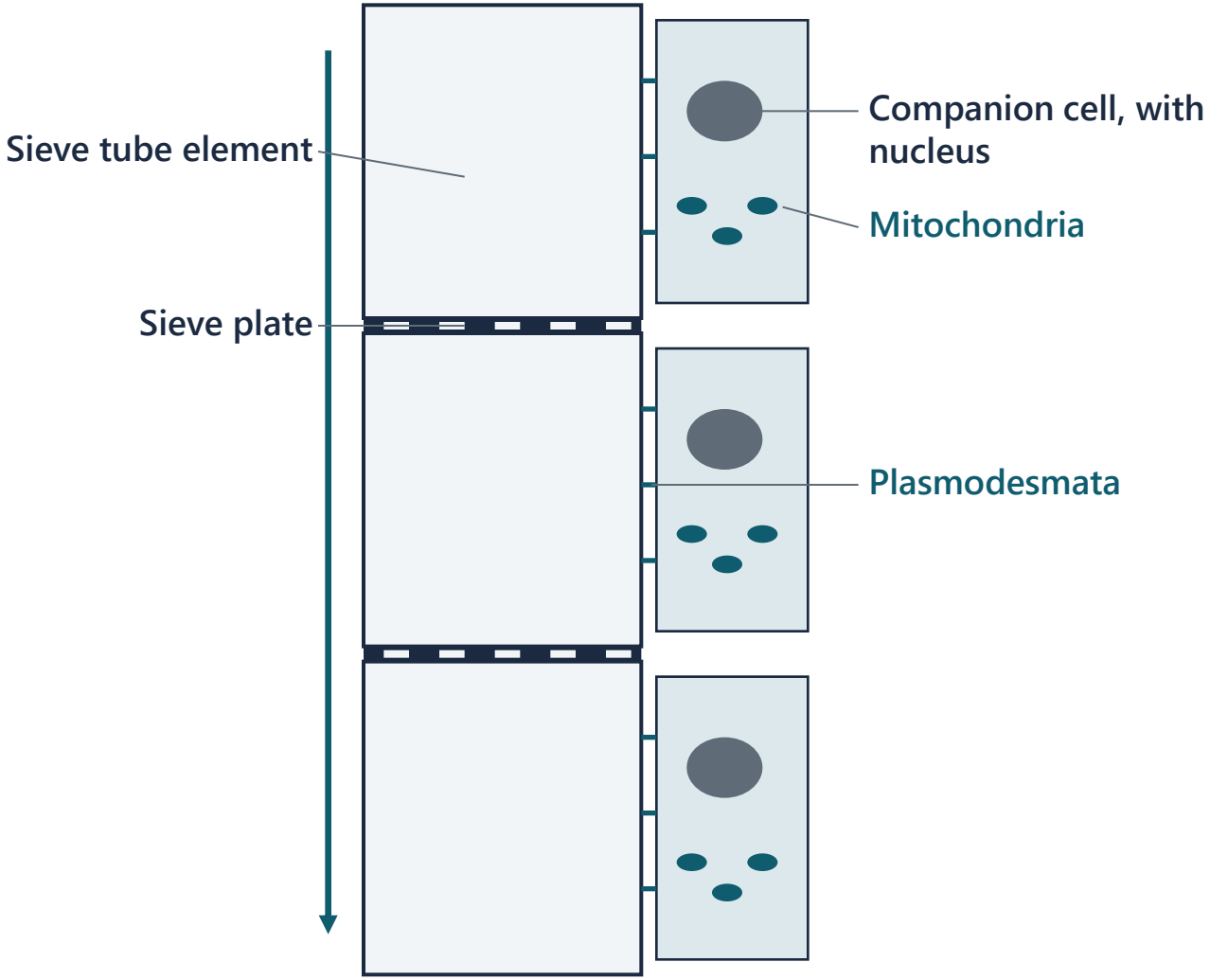
Guttation: drops pushed out of a leaf by root pressure. Public domain.

- Usually water is **pulled** up by transpiration (tension)
- When transpiration is low, roots can **push** sap up instead
- This happens in very humid air, at night, or in spring before deciduous leaves open
- It can refill xylem vessels that have filled with air

Root cells pump mineral ions into the xylem (active transport, ATP) → Xylem sap becomes more concentrated → Water follows by osmosis → Positive pressure pushes sap up the xylem

Sieve tubes and companion cells

B3.2.18



Simplified longitudinal section. The arrow shows sap flow, which can go either way.

- **Translocation**: moving sap with sugars and amino acids from **sources** to **sinks**
- Source: where sugars are made or released, such as mature leaves
- Sink: where they are used or stored, such as roots, fruits and growing buds
- Sap flows under pressure in living sieve tubes

Adaptations for translocation

B3.2.18

Feature	How it helps
Sieve plates with large pores	Sap flows from one element to the next
Little cytoplasm, few organelles	Less resistance to the flow of sap
No nucleus in sieve tube elements	More space for flow; the companion cell keeps the element alive
Many mitochondria in companion cells	ATP for active loading of sucrose at sources
Plasmodesmata between the two cells	Sucrose and other molecules pass into the sieve tube
Thick, rigid cell walls	Withstand the high pressure of the sap

Key idea

Loading at sources raises pressure; unloading at sinks lowers it, so sap flows from source to sink.

Going further: sources and sinks

B3.2.18

Phloem carries sugars from where they are made or stored (sources) to where they are used or stored (sinks). The same organ can change role over the year.

Organ	When it is a source	When it is a sink
Mature leaf	Photosynthesising in summer	Rarely: only when young
Root or tuber	In spring, sending stored sugar to new shoots	In late summer, storing starch
Developing fruit	Never	While growing and ripening

Key idea

Unlike xylem, phloem can carry sap up or down, depending on where the sinks are.

Going further: evidence from aphids and tracers

B3.2.18

- **Aphid stylets**: aphids push their mouthparts into a single sieve tube. If the aphid is cut off, sap keeps flowing out under pressure and can be collected and analysed.
- **Radioactive tracers**: a leaf is given carbon dioxide containing radioactive carbon. Labelled sugars later appear in the phloem and in sinks such as roots and fruits.
- **Ringing**: removing a ring of bark (with the phloem) makes sugars build up above the ring.

Think

Why does sap flow out of a cut aphid stylet, and what does this tell us about the phloem?

Check-in: pushing water and moving sugar

B3.2.17–18

1. Explain how root pressure is generated.
2. State when root pressure is most important.
3. Describe the structure of a sieve tube and its companion cell.
4. Explain how sugar is loaded into the phloem at a source.
5. Explain how pressure differences move sap from source to sink.

How to use this

Answer without your notes, then compare with a partner.

Working with data: a tracer experiment

B3.2.18

Radioactivity in parts of a bean plant 6 hours after one leaf was given labelled carbon dioxide. These data are invented for practice.

Part of plant	Radioactivity / %
Labelled leaf	48
Developing pods	30
Roots	14
Other mature leaves	2
Stem	6

1. Identify the main sinks.
2. Explain why other mature leaves have so little.
3. Explain why the stem has some radioactivity.

Common mistakes to avoid

Not this	But this
Phloem only moves sap downwards.	It moves sap from sources to sinks, in either direction.
Sieve tubes have a nucleus.	Companion cells have the nucleus.
Root pressure is the main way water rises in tall trees.	Transpiration pull is the main force; root pressure helps at times.

Check that you can...

1. **Explain** how root pressure is generated and when it matters
2. **Describe** sieve tubes and companion cells
3. **Explain** translocation from source to sink
4. **Evaluate** evidence for the pressure-flow model

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Outline how active transport in root cells leads to root pressure.

2. State two adaptations of sieve tube elements for the flow of sap.

3. Explain why sap flows from a source to a sink.

Image credits

Helianthus stem.jpg – Jon Houseman – CC BY-SA 4.0

Guttation ne.jpg – NoahElhardt assumed (ba – Public domain

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